Answer Key - Mathematics - K-12 Level Assessment 1 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** C
   Options: A) 96 over 216; B) 16 over 36; C) 4 over 9; D) 9 over 4
   Note: Correct option: C - 4 over 9

2. **Answer:** C
   Options: A) 20; B) 16; C) 9; D) 8
   Note: Correct option: C - 9

3. **Answer:** A
   Options: A) 48,000; B) 480; C) 4,800; D) 5
   Note: Correct option: A - 48,000

4. **Answer:** B
   Options: A) 3^7; B) 7^3; C) $7^4 9$; D) $294^4 9$
   Note: Correct option: B - 7^3

5. **Answer:** C
   Options: A) surveying the cheerleaders for the home team; B) surveying people wearing hats for the visiting team; C) surveying a group of people standing in line for tickets; D) surveying people who do not live in the home team's city
   Note: Correct option: C - surveying a group of people standing in line for tickets

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

7. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

8. **Answer:** False
   Options: A) True; B) False
   Note: Actual answer is '$\frac{1}{8}$'.

9. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

10. **Answer:** True
    Options: A) True; B) False
    Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** The process of choosing a block can be represented by a generating function. Each choice we make can match the 'plastic medium red circle' in one of its qualities (1) or differ from it in k different ways (kx). Choosing the material is represented by the factor $(1 + 1x)$, choosing the size by the factor $(1 + 2x)$, etc:

    $$(1 + x)(1 + 2x)(1 + 3x)^2$$

    Expanding out the first two factors and the square:

    $$(1 + 3x + 2x^2)(1 + 6x + 9x^2)$$

    By expanding further we can find the coefficient of x^2, which represents the number of blocks differing from the original block in exactly two ways. We don't have to expand it completely, but choose the terms which will be multiplied together to result in a constant multiple of x^2:

    $$1 \cdot 9 + 3 \cdot 6 + 2 \cdot 1 = 29$$

    Note: Students should show all steps in their mathematical solution.

12. **Answer:** 1:03. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.
    Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key